Mzuzu University

Department of Mathematics and Statistics

MATH 1101 Precalculus

End of Semester Examination

Monday, March 2, 2020 Time Allowed: 3 hours (2:00-5:00 pm)

INSTRUCTIONS

1. Write the name of your lecturer on the cover of your answer book.
2. If you are repeating the course, write the word "repeater" on the cover of your answer book.
3. Check that the paper has **12** questions.
4. Attempt **all** questions.
5. Answers must be presented in the same order as the questions are presented.
6. Show **all** your working and use appropriate notation.

1. (a) Simplify $\left(\frac{1}{4}\right)^{-2} + \left(\frac{2}{3}\right)^{-2}$. $\boxed{3}$

   (b) Evaluate $\sqrt{\sqrt[3]{64}}$. $\boxed{3}$

2. Rationalise the numerator in the expression $\boxed{5}$

$$\frac{\sqrt{x+h} - \sqrt{x}}{h}.$$

3. Rewrite the following expression without using absolute values given that $x < 3$ $\boxed{3}$

$$\left|x^2 + 1\right| + |x - 3|.$$

4. Determine the center and radius of a circle with the equation $x^2 + y^2 - 10x + 2y + 17 = 0$. $\boxed{5}$

5. Solve the inequality and specify the answer using interval notation. $\boxed{7}$

$$\frac{x+3}{x-4} \geq 0$$

6. Let $f(x) = \dfrac{x+2}{x-3}$.

   (a) Find the domain and range of the function $f(x)$. $\boxed{4}$

   (b) Find $f^{-1}(x)$. $\boxed{3}$

7. Use translations and reflections to sketch the graph of $f(x) = -e^x + 1$. Also specify the domain, range, intercepts and asymptotes of $f(x)$. $\boxed{7}$

8. Show that $\log_b\left(\dfrac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}\right) = 2\log_b(\sqrt{3}+\sqrt{2})$ by using an appropriate rationalization. $\boxed{4}$

9. Solve $3(\ln x)^2 - \ln x^2 - 8 = 0$ for x. $\boxed{6}$

10. Convert

    (a) $450°$ to radians. $\boxed{2}$

    (b) $\dfrac{11\pi}{6}$ to degrees. $\boxed{2}$

11. (a) If $\sec\theta = \frac{7}{5}$ and $270° < \theta < 360°$, find $\tan\theta$. $\boxed{4}$

    (b) Evaluate $\cos\left(-\dfrac{5\pi}{4}\right)$. $\boxed{4}$

12. (a) Simplify the expression $\dfrac{\cot^2\theta}{\csc^2\theta} + \dfrac{\tan^2\theta}{\sec^2\theta}$. $\boxed{4}$

    (b) Prove the identity $\sin A + \cos A = \dfrac{\sin A}{1 - \cot A} - \dfrac{\cos A}{\tan A - 1}$. $\boxed{4}$